



IIT PATNA
VISHLESAN I-HUB
FOUNDATION



Baroda Institute
of Technology

CAPSTONE PROJECT-1

Exploratory Data Analysis

HOTEL BOOKING ANALYSIS

Abstract: Our data set contains booking information for a city hotel and a resort hotel.

It includes information such as when the booking was made, length of stay, the number of adults, children or babies, cancellation count, revenue and other things.

In upcoming slide, we will analysis and get insight from the data.



POINTS FOR DISCUSSION

- Descriptive summary
 - Hotel wise analysis
- Distribution Channel wise analysis
 - Booking cancellation analysis
 - Timewise analysis
- Relation between deposit type and cancelation
 - Leading Hotel
 - Waiting time
- Ordered meal at the hotel
 - Hotel Revenue
- ADR – Monthly & Highest ADR

PYTHON LIBRARIES USED

- NumPy
- Pandas
- Seaborn
- Matplotlib
- Plotly lib



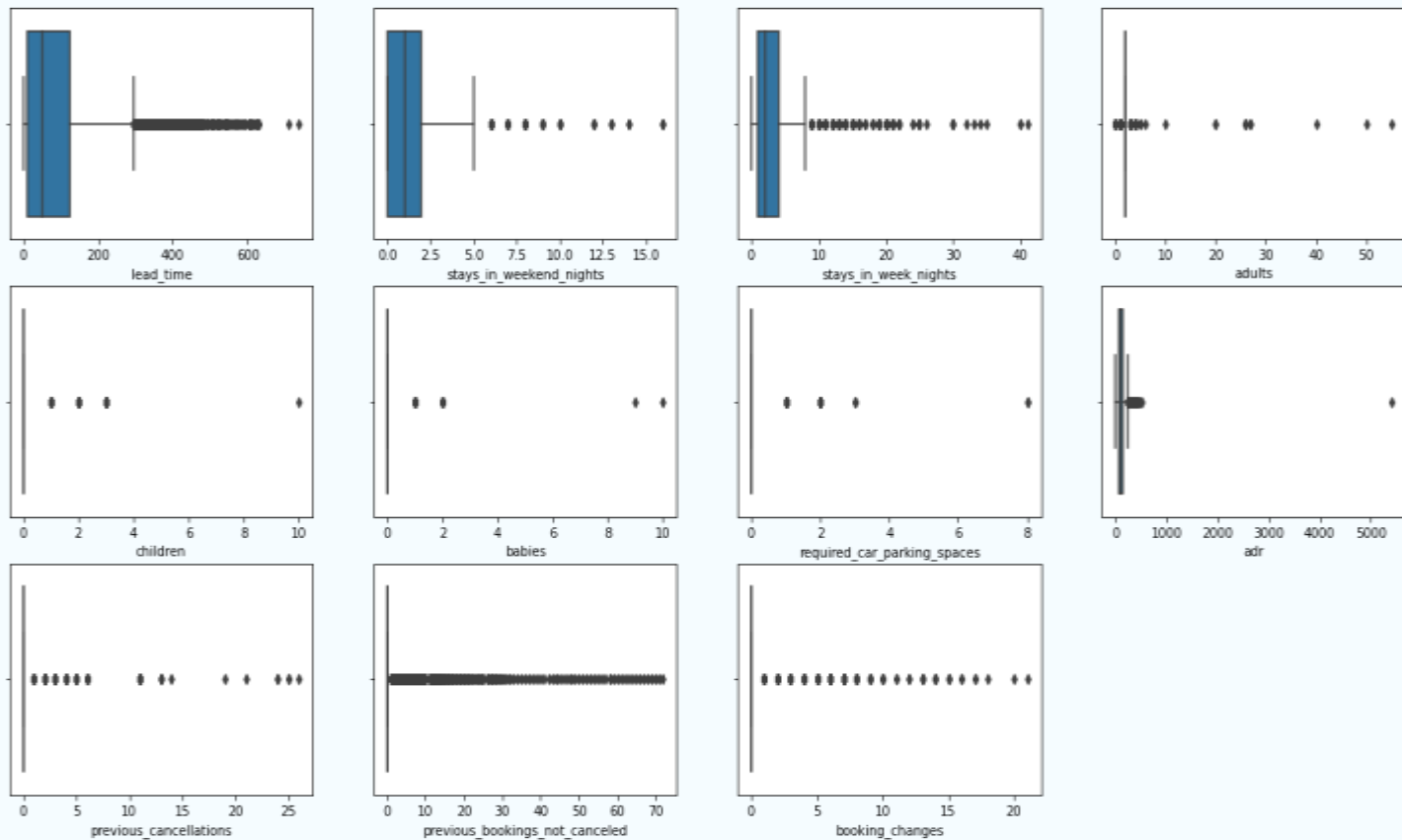
seaborn



PREPARING OUR DATA FOR DEEP ANALYSIS

- Step 1. Overview the whole data : number of rows and columns, D-type(Data type), null rows.
- Step 2. Looking at missing data
- Step 3. Splitting the dataset into numerical and categorical(String)
- Step 4. Look at data distribution for numerical columns
- Step 5: Look at Data Characteristics of text variables
- Step 6. Looking at the duplicate & null values and dropping them.
- Step 7: Conclusion

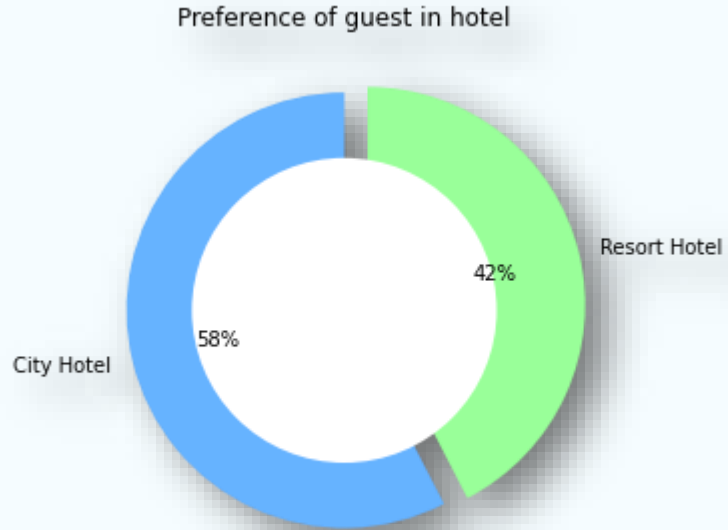
OUTLIERS



MOST PREFERRED HOTEL

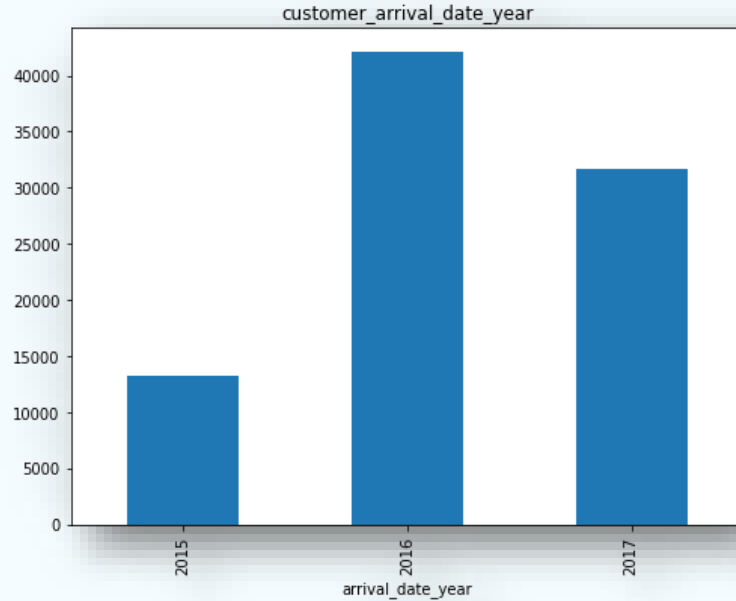
City hotels are more preferred by customer as compare to resort hotel.

61% customers booked for city hotel **and 39%** booked for resort.

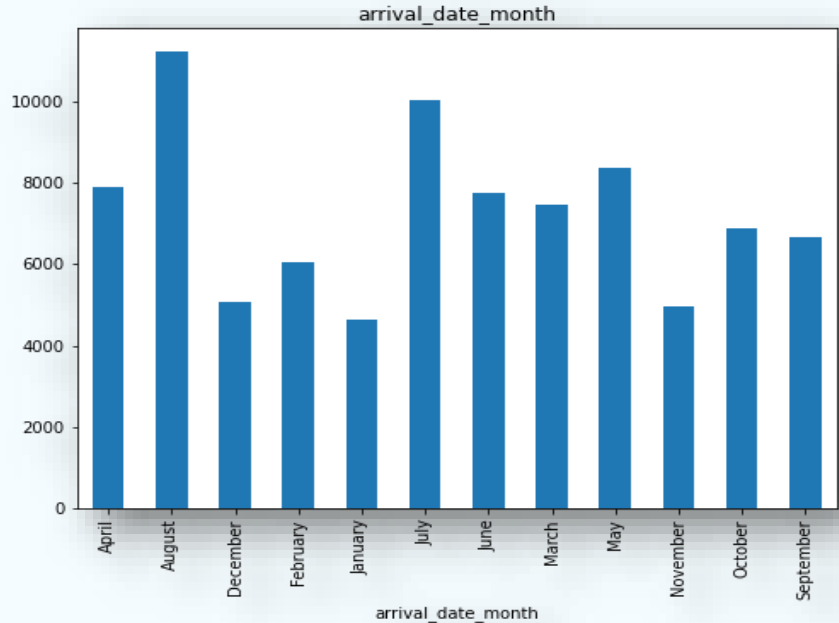


CUSTOMER ARRIVAL

Yearly

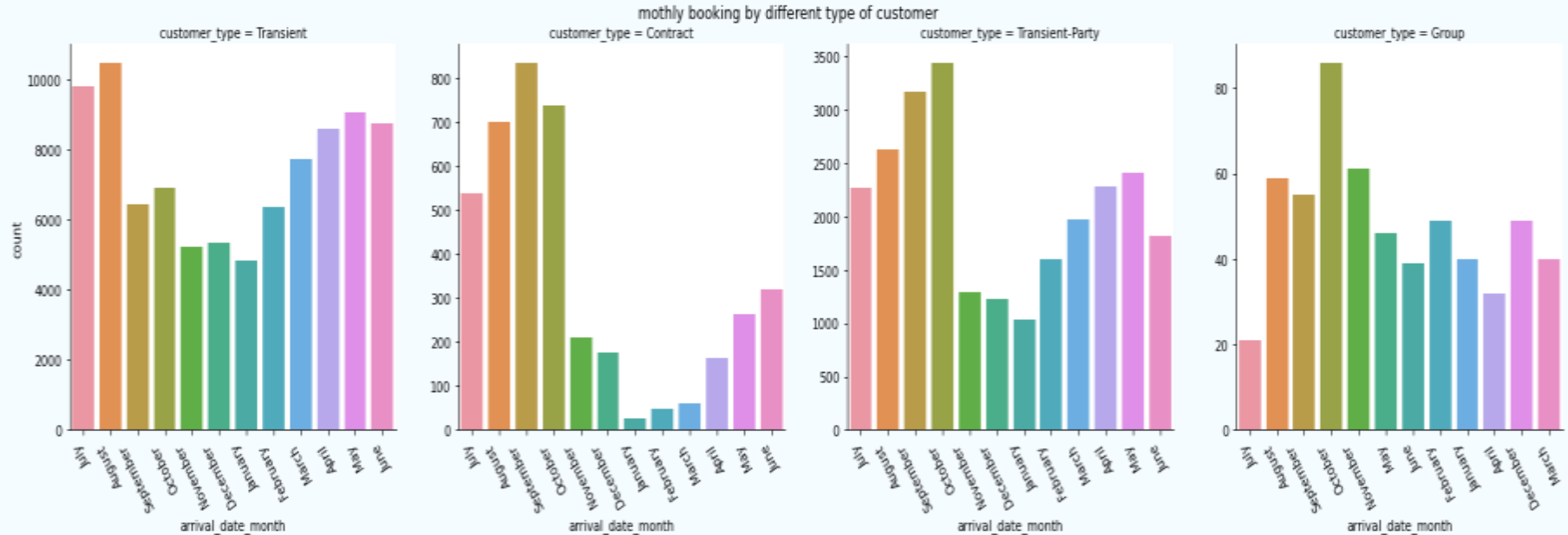


Monthly



Maximum customer arrive in to the hotel in year 2016.
In august month, maximum number people checked in.

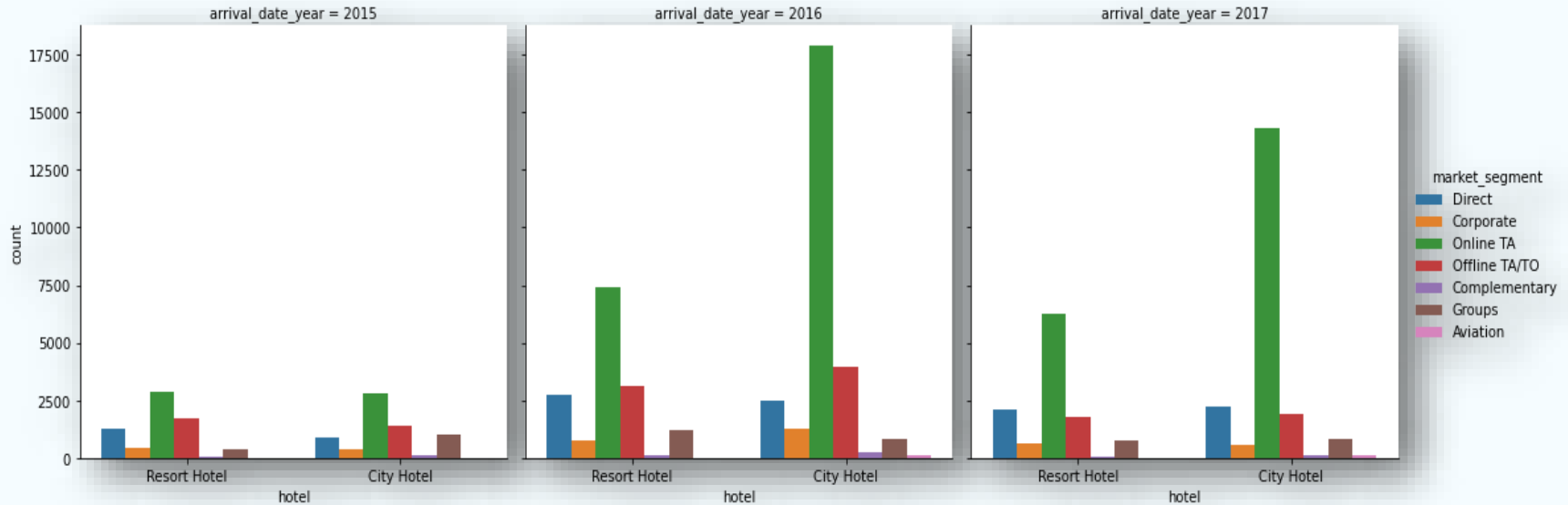
ANALYSIS BASED ON MARKET SEGMENT



Here, we can see the most active month as indicated by the appearance of each type of customer.

Highest bookings - Transient customer booked for **august** month, **contract** booked for **September**, **transient party and group** both booked for **October** month.

ANALYSIS BASED ON MARKET SEGMENT



Aviation and complementary bookings are negligible compared to the **online and offline TA**

Corporate ,direct and group bookings together add up to **online TA**.

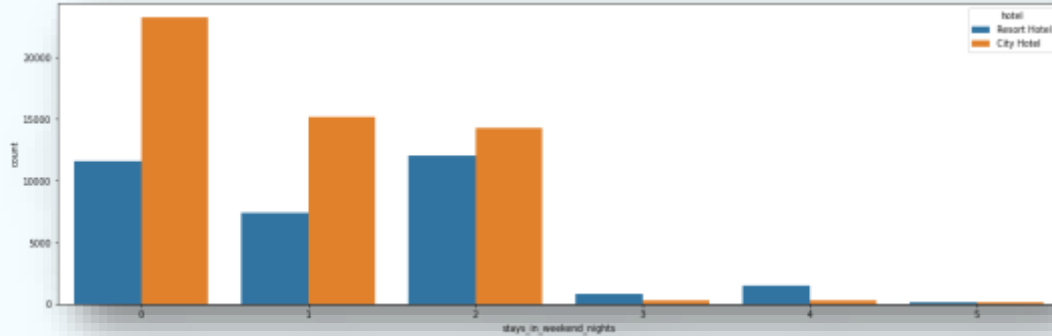
COUNTRY RECORD



Most of the customers from European countries like Portugal, Great Britain, France and Spain. Around 40% of all bookings were booked from Portugal followed by Great Britain(10%) & France(8%).

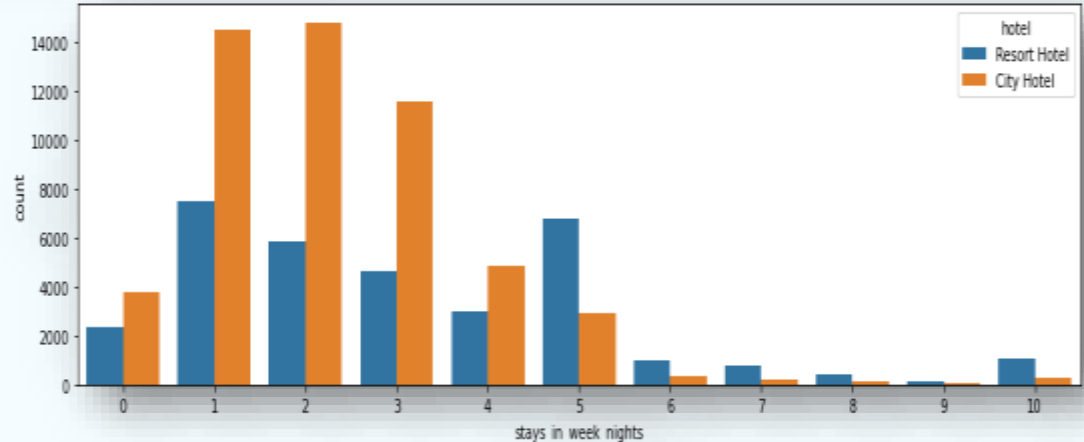
CUSTOMER STAYS

Weekend Nights

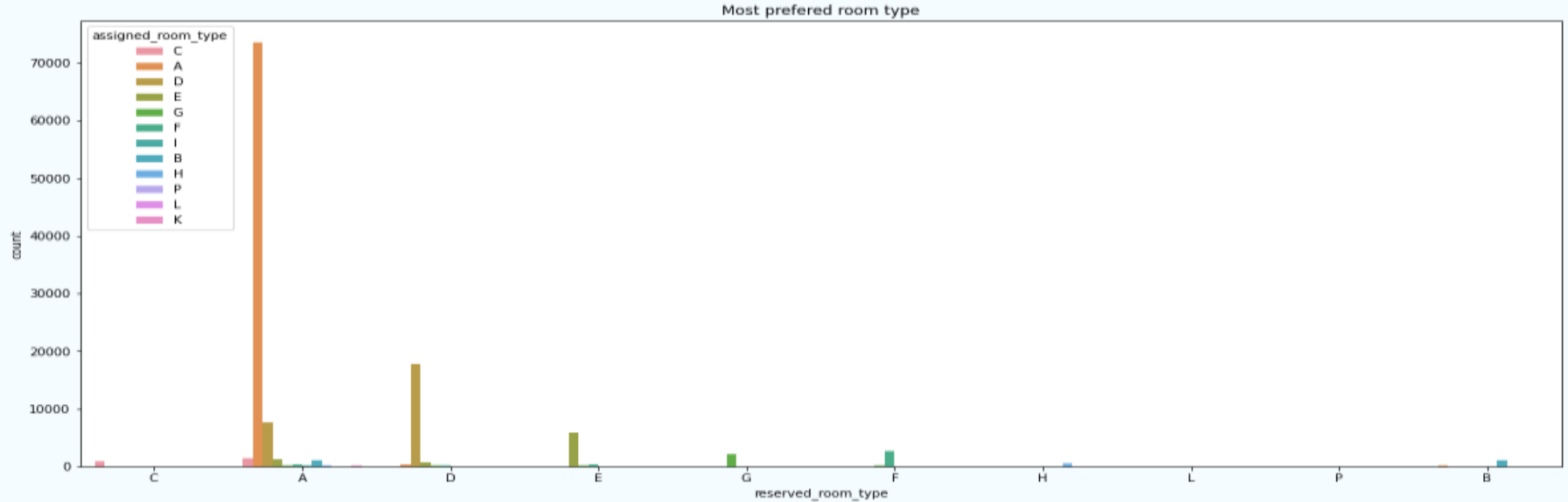


We observe that weekday night stays were more in City Hotel as compare to weekend night.

Weekdays Nights



ROOMS ASSIGNMENT

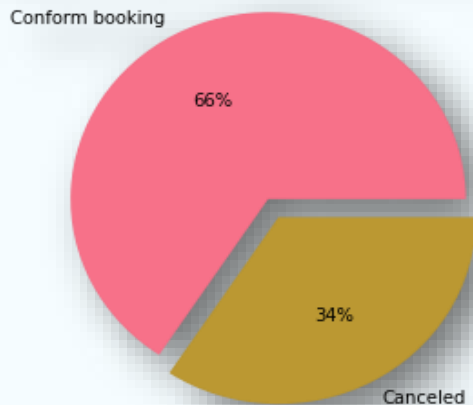


A- type of customer assigned same room as they reserved followed by D-type and E-type.

CANCELLATION

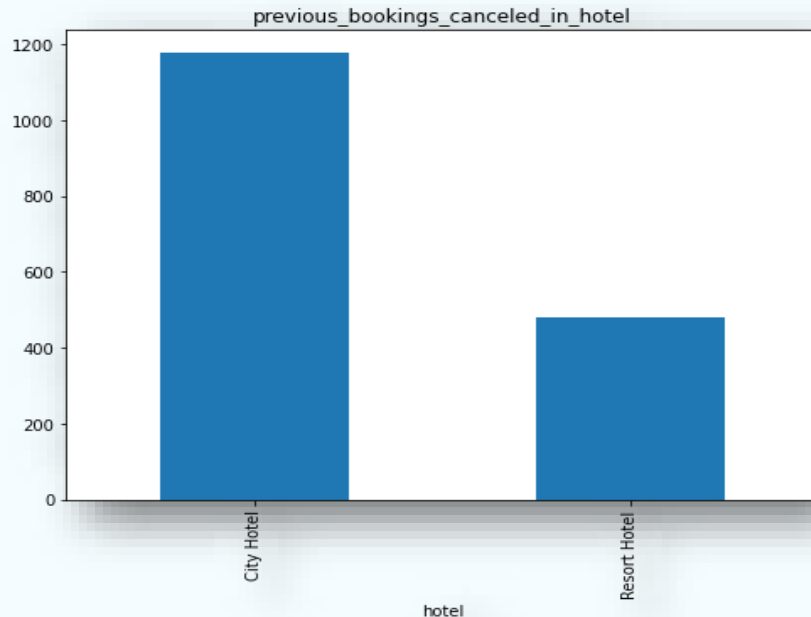
Ratio between cancelled and confirmed bookings?

Percentage ratio between cancelled and confirmed bookings



Confirmed bookings are slightly more than the cancelled bookings.

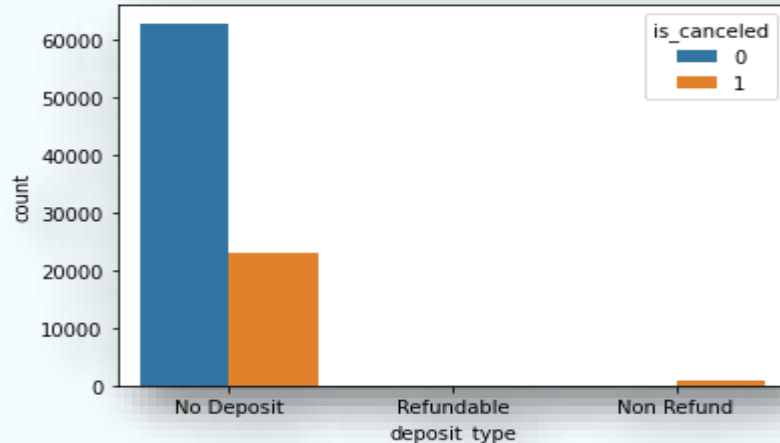
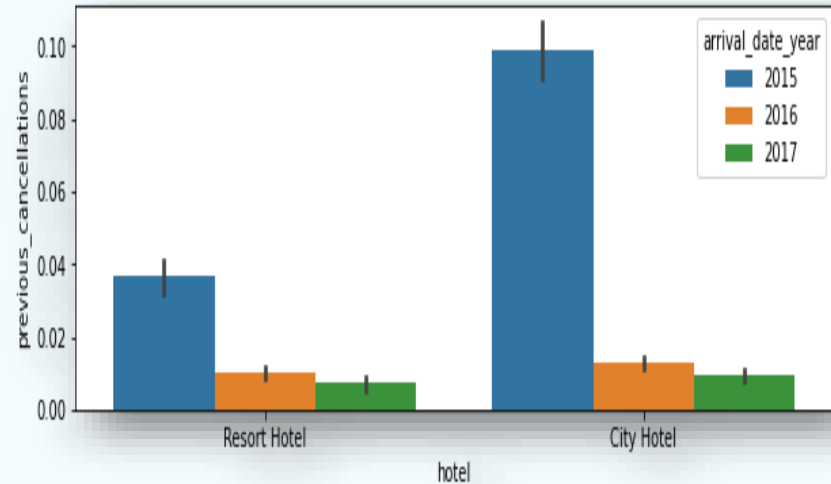
Highest cancellation between hotels?



City hotel has higher cancellation than the resort hotel.

Cancellation with arrival date-year of hotel respectively?

In year 2015, the count of cancellation is higher than 2016 and 2017.
That too from city hotel.

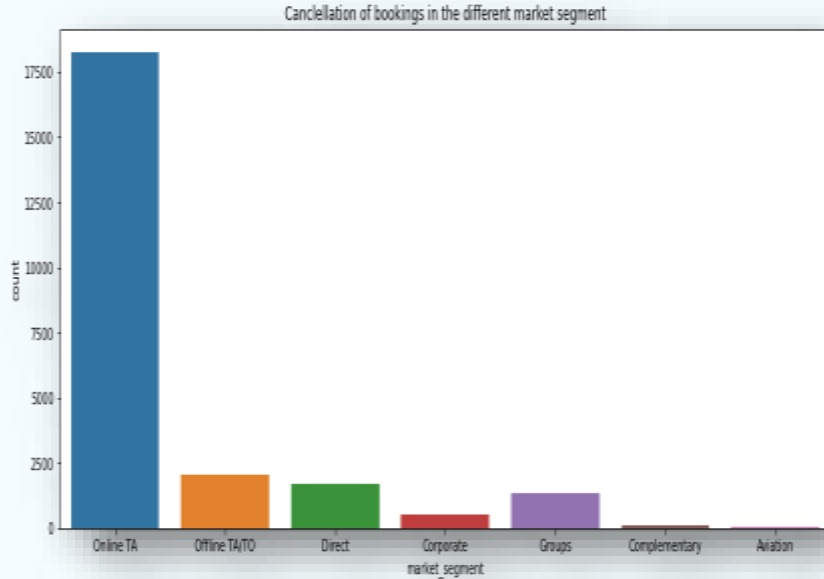


Relation between deposit type and cancellation?

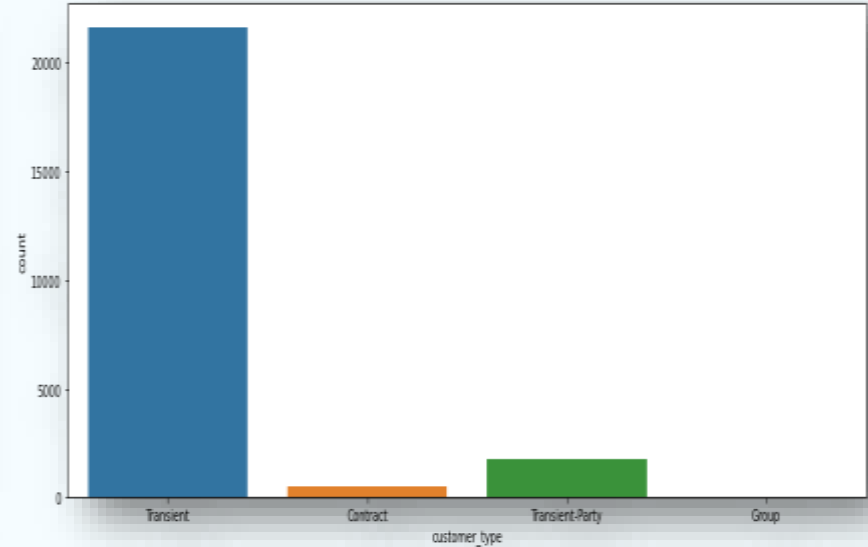
Refundable is least type of preferred deposit amongst all
Cancellation is higher in hotel which offers no deposited/no advance payment.

CANCELLATION ANALYSIS BASED ON:

Market Segment

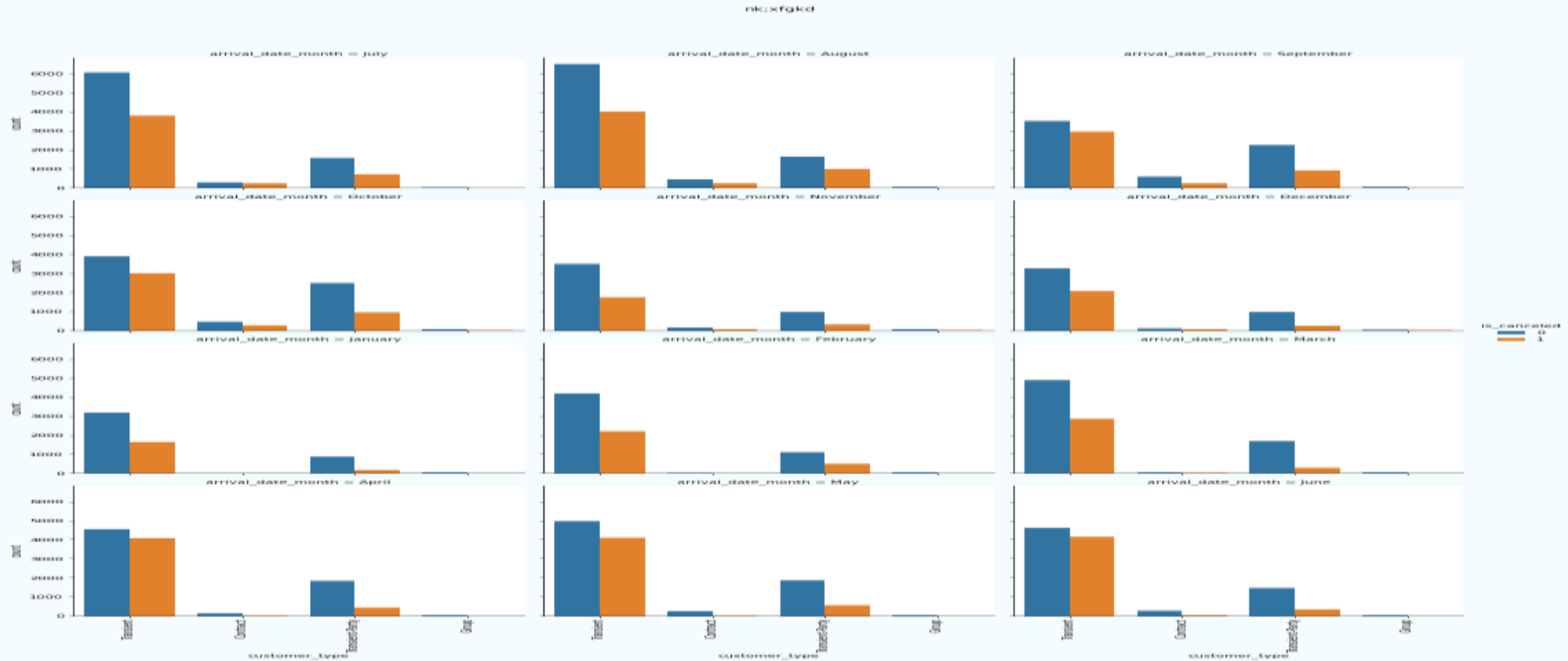


Customer Type



Transient customer from online booking platform had cancelled their booking more frequently.
Aviation and complementary bookings are negligible compared to the online and offline TA

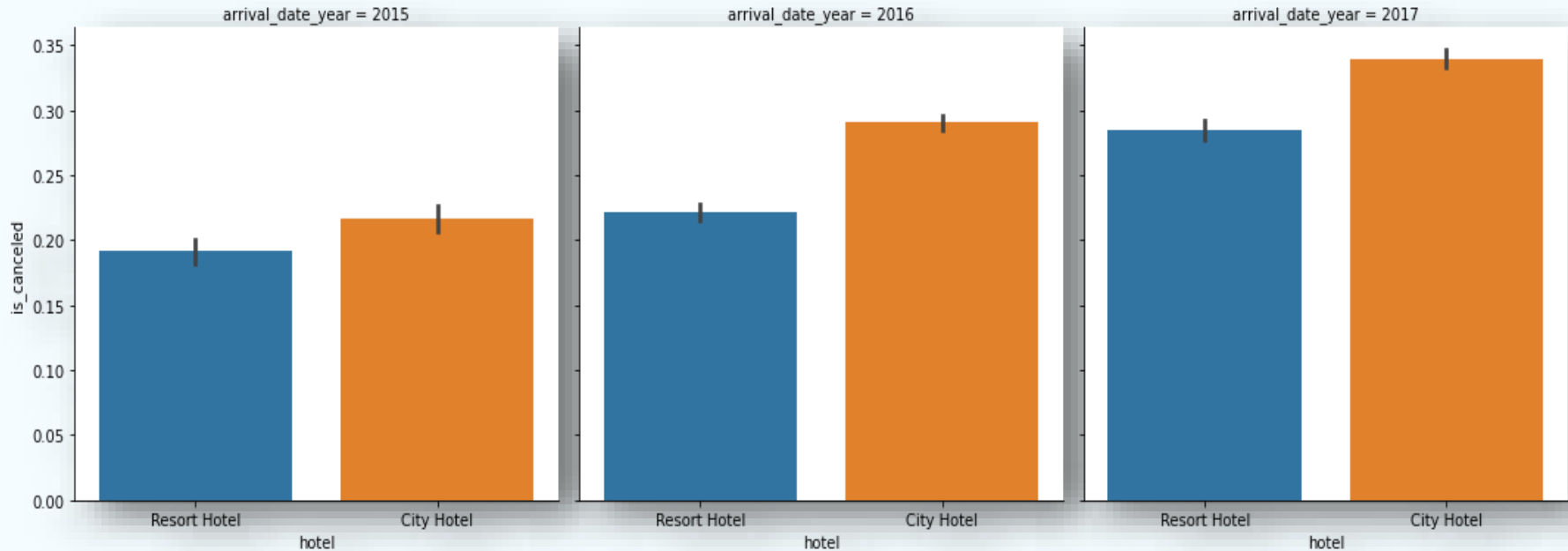
CANCELLATION: MONTHLY ANALYSIS



Cancellation rate is higher by the transient customer in the month of august.

Although November and December has almost same bookings ,cancellation rate in December is more

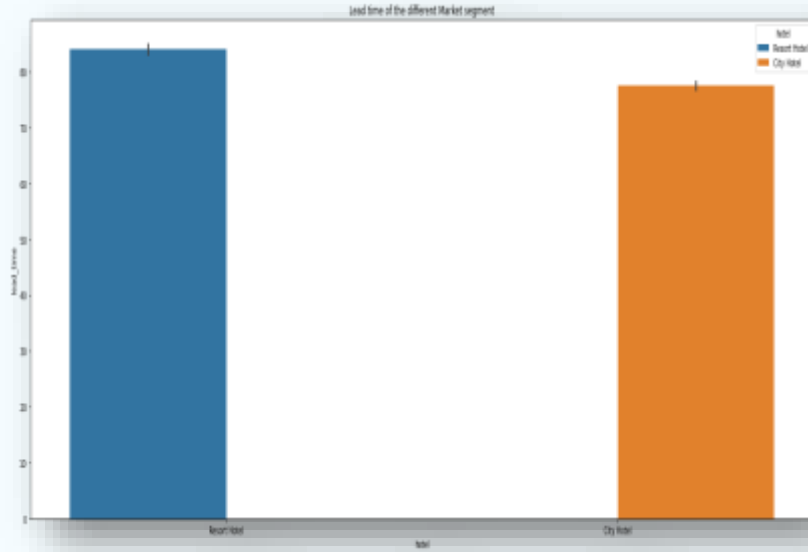
REPEAT CUSTOMER



Maximum repeat customers were arrived in city hotel.

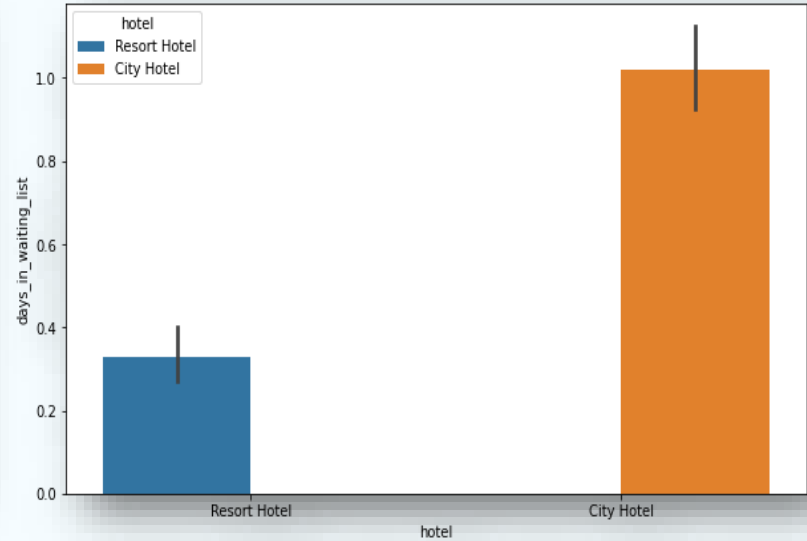
In year 2017, maximum repeat customer were noticed.

LEADING TIME



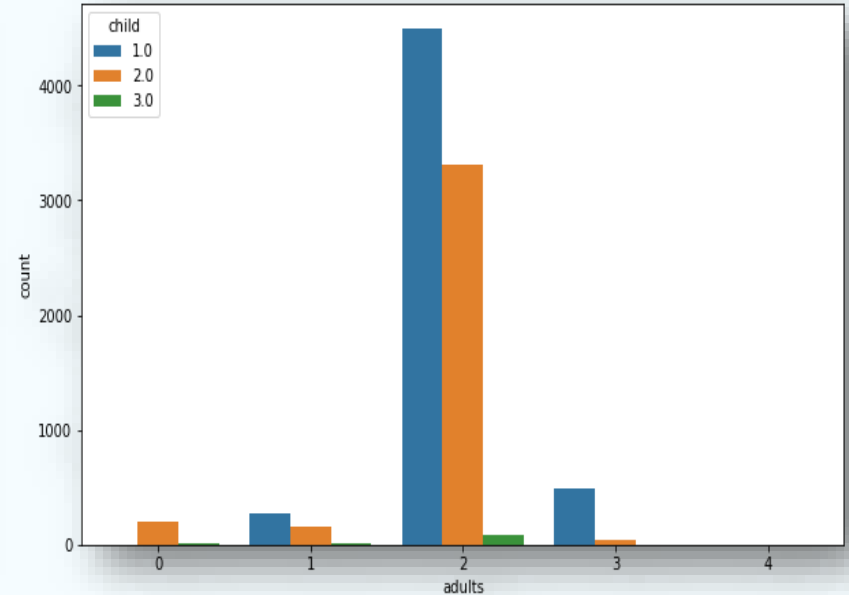
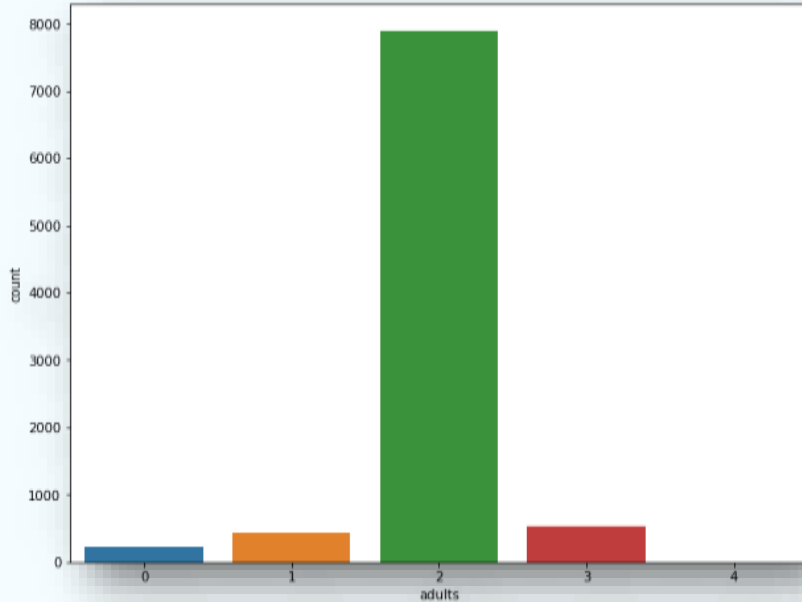
Resort hotel has higher lead time over the city hotel.

WAITING TIME



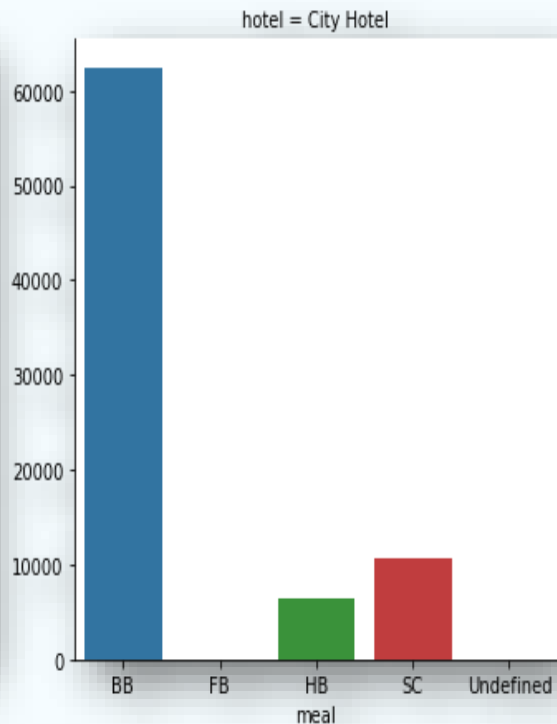
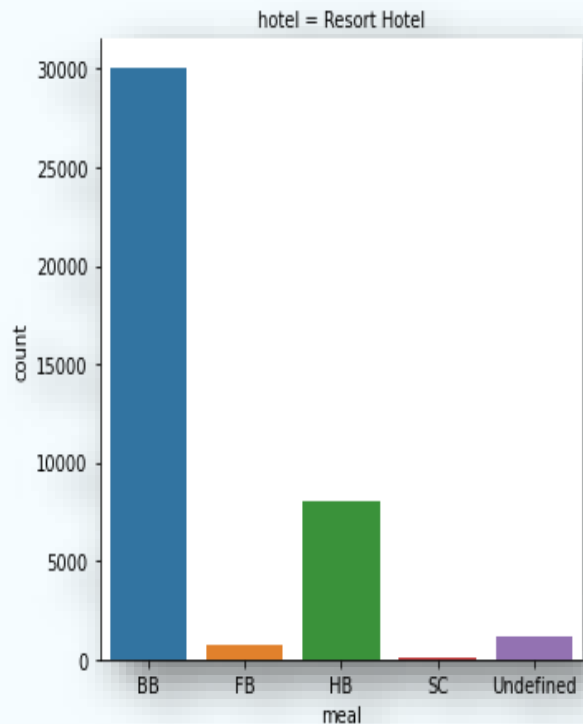
City hotel has longer waiting time than resort hotel.

GUEST ARRIVAL: ADULTS & KIDS



Most visitors were arrived in pair with no children/babies and preferred city hotel over resort hotel.
Visitors who had 1 or 2 children also preferred city hotel

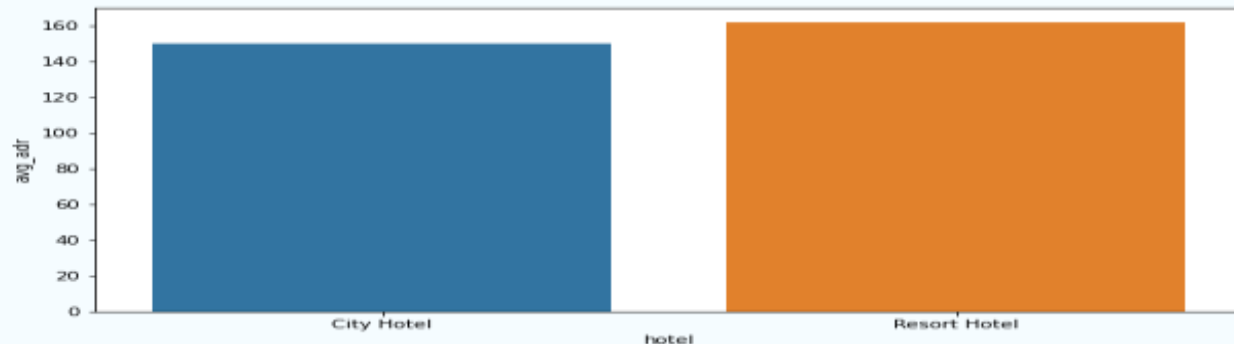
MOST ORDERED MEAL AT THE HOTEL



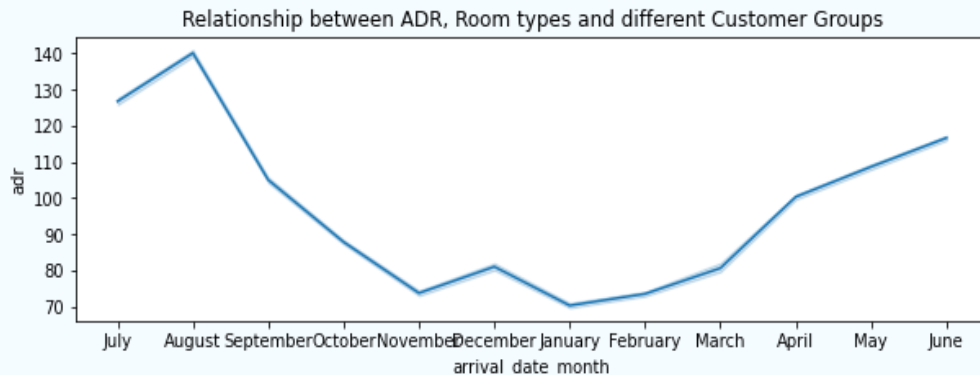
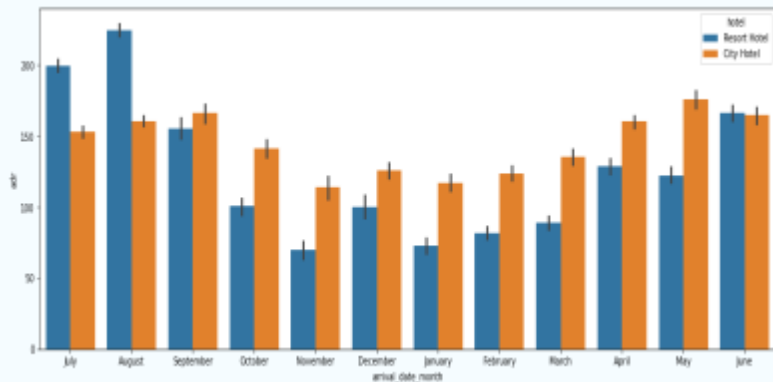
Out of the meals, BB (Bed & Breakfast) is the most ordered meal which is around 77.2%, followed by HB(Half Board), SC(no meal package), Undefined and FB (Full Board).

REVENUE

Resort hotel has generated more revenue as compare to city hotel.



HIGHEST ADR



Highest ADR is observed in august month. However resort's ADR is highest in august and city hotel ADR is highest in may month

CONCLUSION

- There are more city hotel than resort hotel. Booking rate is also higher in city hotel.
- August is the most demanded month and January is the least. Hotel management should come up with interesting offers/deals in the months when they have least booking
- Repeat customers are more in city hotel and the number is increasing every month. Resort management needs to improve the hotel facilities to make more repeated guest.
- Resort hotel is likely to make more revenue.
- We have seen that most of the guests were from western Europe countries. Portugal is the most active country.
- Since, cancellation is higher is online-ta. Hotel management should work on their terms & policies and taking feedback seriously for improvement.

THANK YOU!